

Galileo observations of volcanic plumes on Io

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ABSTRACT

Io's volcanic plumes erupt in a dazzling variety of sizes, shapes, colors and opacities. In general, the plumes fall into two classes, representing distinct source gas temperatures. Most of the Galileo imaging observations were of the smaller, more numerous Prometheus-type plumes that are produced when hot flows of silicate lava impinge on volatile surface ices of SO₂. Few detections were made of the giant, Pele-type plumes that vent high temperature, sulfur-rich gases from the interior of Io; this was partly because of the insensitivity of Galileo's camera to ultraviolet wavelengths. Both gas and dust spout from plumes of each class. Favorably located gas plumes were detected during eclipse, when Io was in Jupiter's shadow. Dense dust columns were imaged in daylight above several Prometheus-type eruptions, reaching heights typically less than 100 km. Comparisons between eclipse observations, sunlit images, and the record of surface changes show that these optically thick dust columns are much smaller in stature than the corresponding gas plumes but are adequate to produce the observed surface deposits. Mie scattering calculations suggest that these conspicuous dust plumes are made up of coarse grained "ash" particles with radii on the order of 100 nm, and total masses on the order of 10⁶ kg per plume. Long exposure images of Thor in sunlight show a faint outer envelope apparently populated by particles small enough to be carried along with the gas flow, perhaps formed by condensation of sulfurous "snowflakes" as suggested by the plasma instrumentation aboard Galileo as it flew through Thor's plume [Frank, L.A., Paterson, W.R., 2002. *J. Geophys. Res. (Space Phys.)* 107, doi:10.1029/2002JA009240, 31–1]. If so, the total mass of these fine, nearly invisible particles may be comparable to the mass of the gas, and could account for much of Io's rapid resurfacing.

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1. Introduction

Io's explosive volcanic eruptions are among the most spectacular but least well understood phenomena in the Solar System. These violent eruptions take place when volatile vapors are expelled from volcanoes at speeds of up to 1000 m per second, forming plumes of gas and dust that reach hundreds of kilometers above the satellite's surface. Only a few of Io's many active volcanoes have been observed to produce plumes, and another handful of volcanic centers have been inferred to have plume activity through observations of the surface changes caused by deposits emplaced by plumes. Several of Io's plumes have apparently been sustained for decades, while others erupted in episodic big bangs that lasted less than a few weeks.

Voyager and Galileo observations have established that there are two classes of plumes on Io (McEwen and Soderblom, 1983;

Geissler et al., 2004a): giant plumes like Pele¹ that vent sulfur-rich gases from the interior of the moon and spray-paint the surface with enormous red rings, and more numerous smaller plumes like Prometheus that are produced when hot flows of silicate lava impinge on volatile surface ices of SO₂. Both classes of plumes possess gas and dust components, although the density of dust and gas varies from one plume to another, even within the same class. Fallout from the plumes continually coats the surface in a constantly changing variety of colors, while the gases contribute substantially to Io's tenuous atmosphere. A fraction of the dust and gas ejected by plumes is swept away by Jupiter's magnetic field, helping to sustain the neutral clouds, plasma torus, and dust streams escaping Io.

Io's plumes were intensively studied soon after their discovery in Voyager imaging data. Strom and Schneider (1982) provided detailed observational descriptions of the plumes that were imaged by Voyagers 1 and 2. Collins (1981) estimated the dust

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¹ Due to their variability in space and time, plumes on Io are not given names. It should be understood that references to plumes are in fact references to the specific plumes issuing from the corresponding named eruptive centers.

particle size and mass of Loki's dust plume by fitting a Rayleigh scattering law to Voyager 1 color observations. Kieffer (1982) presented a thorough theoretical treatment of the thermodynamics of plumes and their possible sources. Johnson and Soderblom (1982) pointed out the importance of plumes for global resurfacing and heat flow. McEwen and Soderblom (1983) studied plume deposits and surface changes between the Voyager 1 and 2 encounters and recognized the distinction between the two classes of plumes on Io.

However, many aspects of the plumes remained mysterious until the arrival of Galileo in 1995. Still poorly known are the temporal and spatial variability of plumes, the relationship between the plumes and the surface changes they produce, and the nature and sources of the particulates present in plumes of both classes. Among the outstanding unanswered questions are the following:

- How do the plumes alter the surface of Io? Geissler et al. (2004a) documented dozens of surface changes that took place during the 5 year tenure of Galileo, and found surface changes at every location in which plumes had been spotted. Are the visible surface changes due to dust deposition from the plumes, or to condensation of gas from the plumes onto the surface?
- How are dust particles formed? Does it “snow” sulfurous flakes that are condensed as the gas expands and cools? Or are the dust particles made up of volcanic ash that is entrained with the gas flow?
- How big are the dust particles? Estimates of the dust particle radii in Io's plumes range from <10 nm (Collins, 1981) to 0.1 mm (James and Wilson, 1998), depending upon the assumptions made, an inconsistency of 4 orders of magnitude.
- How massive are the dust plumes? Published estimates of dust production rates for Io's plumes also vary greatly, ranging from 10^3 kg s $^{-1}$ for Pele (Spencer et al., 1997) through 10^7 kg s $^{-1}$ for both Pele and Pillan (Cataldo et al., 2002), up to 10^8 kg s $^{-1}$ for Loki (Collins, 1981).
- How significant are volcanic plumes to Io's resurfacing? Answers to the previous questions are needed to determine whether the plumes can account for the rapid resurfacing of Io, responsible for erasing any hint of bombardment of the moon by impactors. A globally averaged resurfacing rate of ~ 1 mm yr $^{-1}$ is required in order to account for the burial of impact craters on Io (Spencer and Schneider, 1996), equivalent to the emplacement of 1000 m 3 s $^{-1}$. How much of this is due to plume deposits and how much to the emplacement of silicate lava flows is an open question.

In this article, we begin to address the above problems through a systematic study of the dimensions, colors and opacities of dust plumes in the complete set of Galileo images, together with comparisons of the dust plumes seen in daylight to the gaseous auroral glows seen in eclipse and the record of surface changes. In the following sections we first outline our measurement methodologies before presenting the Galileo data on the sizes and morphologies of dust and gas plumes, and the colors and scattering behavior of the Prometheus-type dust plumes for which numerous observations exist. Next we interpret the color data using Mie theory to derive constraints on dust particle sizes, and apply these constraints to estimate the masses of the dust plumes and the corresponding mass production rates. Finally we discuss the implications of the results for the genesis of dust in Io's plumes, revisit the importance of the plumes for global resurfacing, outline the deficiencies in our knowledge of Io's plumes, and suggest directions for further research.

2. Approach

Galileo Solid State Imaging (SSI) pictures of Io's plumes were acquired with a mix of different geometries, resolutions, phase angles, and compression ratios. Many of these images were captured in targeted campaigns specifically intended to monitor plume activity on Io (McEwen et al., 1998; Keszthelyi et al., 2001; Turtle et al., 2004). To analyze these data, the images were first imported into the USGS ISIS software system (Gaddis et al., 1997), navigated (camera pointing corrected), and radiometrically calibrated using the procedures of Klaassen et al. (1997) to either reflectance units (I/F) or spectral radiance (nW cm $^{-2}$ sr $^{-1}$ nm $^{-1}$). Individual frames of the color observations were coregistered and assembled into color cubes so that all of the filters could be measured simultaneously.

The heights of dust columns were measured in sunlit images of plumes, correcting for the foreshortening that occurs when the plume is in front of or behind the limb. The correction was calculated by first finding the angle by which the plume was rotated around from the limb using the cosine formula of spherical trigonometry with the known latitudes and longitudes of the spacecraft point and the source of the plume, where the plume source location was based on the tabulation of surface changes from Geissler et al. (2004a). Then a plane cosine correction was applied to derive the true dust column heights from the measured heights.

The heights of gas plumes were measured in eclipse images with a similar foreshortening correction applied. Galileo images of Io in eclipse show many different diffuse glows associated with various components of the atmosphere and electrical currents stimulating the emissions (Geissler et al., 1999, 2001a, 2004b). Only very massive or favorably located plumes (near the sub-Jupiter and anti-Jupiter points of Io) produce visible auroral emissions that could be detected by Galileo's SSI camera. For this study, we measured the heights of gas plumes if they met two criteria: (1) distinct knots of emission with discernable plume morphology, and (2) association with a clearly identifiable volcanic center known to have produced surface changes.

The brightness and color of the sunlit dust plumes were measured as functions of phase angle and corrected for residual calibration errors by subtracting the average values of the dark space background surrounding the plumes. The dust plume brightness (I/F) is strongly affected by the position of the plume with respect to Io's limb, and the sparse Galileo data do not allow us to reliably quantify the scattering phase function of the dust. However, the color observations are self-consistent when normalized to the Violet filter brightness, and these measurements are discussed in Section 3.7.

The total optical power scattered by the sunlit dust plumes was estimated as the product of the mean measured spectral radiance (nW cm $^{-2}$ sr $^{-1}$ nm $^{-1}$) times the solid angle per pixel (1.03×10^{-10} sr for Full Frame mode, 4.13×10^{-10} sr for Summation mode), times the number of pixels subtended by the plume, times the filter bandwidth (45 nm for Violet images; 65 nm for Green).

3. Observations

Almost all of the Galileo plume sightings were of the smaller, more common Prometheus-type plumes. Only two images detected giant, Pele-type plumes, and these are discussed separately in Section 3.5. All of the Galileo plume images were acquired at low resolution (>1.6 km/pixel), comparable to the pictures of plumes taken by the Voyagers. An attempt to image Prometheus at 45 m/pixel during orbit I31 unfortunately failed due to a camera anomaly (Turtle et al., 2004). Because of severe telemetry

Table 1
Dust plume heights

Name	Type	Lat.	Lon.	Orbit	Image	Filter	GMT (M/D/Y H:M)	Res. (km/pix)	Meas. height (km)	Meas. error (km)	Corr. height (km)	Corr. error (km)	Position/notes
Grian	Pe	−12	13	C10	s0413791600	Red	9/19/1997 20:49	10.5	84.2	30	144	51	Limb
Kanehekili	Pr	−19	34	G8	s0394435001	Violet	5/6/1997 22:52	13.0	70.2	13	75	14	Limb
Kanehekili	Pr	−16	36	E11	s0420900323	Violet	11/8/1997 18:45	19.5	100.7	20	101	20	Limb
Masubi	Pr	−44	54	C21	s0506406118	Violet	7/2/1999 4:04	2.6	82.2	20	87	21	Limb
Masubi	Pr	−44	54	C21	s0506406153	Violet	7/2/1999 4:05	3.3	91.0	10	96	11	Limb
Masubi	Pr	−44	54	C21	s0506406800	Green	7/2/1999 4:11	1.6	74.9	10	80	11	Limb
Masubi	Pr	−44	54	C21	s0506584123	Violet	7/3/1999 10:04	16.7	93.9	20	99	21	Limb
Masubi	Pr	−44	54	C22	s0512352501	Violet	8/12/1999 22:09	10.8	63.1	20	64	20	Limb
Masubi	Pr	−44	54	C22	s0512375600	Violet	8/13/1999 2:02	12.8	64.0	20	92	29	Limb
Masubi	Pr	−44	54	I31	s0615693301	Violet	8/7/2001 12:58	19.6	61.4	20	91	30	Limb
Masubi	Pr	−44	54	I31	s0615816301	Violet	8/8/2001 9:42	19.9					Terminator
Amirani	Pr	21	112	C22	s0512420523	Violet	8/13/1999 9:37	15.9	36.6	15	66	27	Limb
Amirani	Pr	21	112	C22	s0512436223	Violet	8/13/1999 12:15	16.4	73.6	15	76	15	Limb
Amirani	Pr	21	112	C22	s0512447723	Violet	8/13/1999 14:12	16.5	36.8	20	87	47	Limb
Amirani	Pr	23	115	E4	s0374850023	Violet	12/20/1996 10:27	18.3	40.4	20	72	36	Limb
Amirani	Pr	24	116	G8	s0394478123	Violet	5/7/1997 6:09	9.8	52.8	15	60	17	Limb
Thor	Pr	39	131	I31	s0615325147	Violet	8/4/2001 22:56	18.3	103.3	18	107	19	Limb
Thor	Pr	39	131	I31	s0615325547	Violet	8/4/2001 23:00	18.2	120.0	10	123	10	Limb
Prometheus	Pr	−1	155	G2	s0359402500	Violet	9/2/1996 23:15	30.8					Terminator
Prometheus	Pr	−1	155	G2	s0359653300	Violet	9/4/1996 17:31	21.9					Terminator
Prometheus	Pr	−1	155	G2	s0359722942	Violet	9/5/1996 5:15	18.7	56.5	20	86	30	Limb
Prometheus	Pr	−1	155	G2	s0359729642	Violet	9/5/1996 6:23	19.0	76.1	20	76	20	Limb
Prometheus	Pr	−1	155	G2	s0359736542	Violet	9/5/1996 7:33	19.3	38.9	20	67	35	Limb
Prometheus	Pr	−1	155	C3	s0368558239	Clear	11/6/1996 6:10	3.5					Disk
Prometheus	Pr	−1	155	E4	s0374575922	Violet	12/19/1996 12:15	11.7					Disk
Prometheus	Pr	−1	155	E6	s0383600826	Violet	2/19/1997 21:07	10.9					Terminator
Prometheus	Pr	−1	155	G8	s0394435001	Violet	5/6/1997 22:52	13.0					Terminator
Prometheus	Pr	−1	155	G8	s0394505123	Violet	5/7/1997 10:41	10.0	89.9	10	90	10	Limb
Prometheus	Pr	−1	155	C9	s0401785407	Violet	6/27/1997 13:33	8.3	66.4	16	71	17	Limb
Prometheus	Pr	−1	155	C9	s0401863204	Violet	6/28/1997 2:40	6.2	295.7	60	74	15	Disk
Prometheus	Pr	−1	155	C9	s0401876300	Violet	6/28/1997 4:52	12.7					Terminator
Prometheus	Pr	−1	155	E11	s0420730085	Violet	11/7/1997 14:05	9.3	68.2	10	79	12	Limb
Prometheus	Pr	−1	155	E11	s0420789285	Violet	11/8/1997 0:03	8.0					Terminator
Prometheus	Pr	−1	155	C21	s0506492200	Violet	7/2/1999 18:34	10.0					Terminator
Prometheus	Pr	−1	155	C22	s0512466924	Violet	8/13/1999 17:26	16.3	48.9	20	72	29	Limb
Prometheus	Pr	−1	155	C22	s0512475823	Violet	8/13/1999 18:56	16.1	58.5	15	59	15	Limb
Zamama	Pr	18	171	G1	s0349673978	Violet	6/26/1996 15:49	13.9	75.3	15	79	16	Limb
Zamama	Pr	18	172	G8	s0394519124	Violet	5/7/1997 13:03	10.5	61.0	20	62	20	Limb
Zamama	Pr	19	175	E11	s0420730085	Violet	11/7/1997 14:05	9.3	29.6	10	80	27	Limb
Zamama	Pr	19	175	E11	s0420743485	Violet	11/7/1997 16:20	9.0	66.1	20	67	20	Limb
Zamama	Pr	19	175	E11	s0420789285	Violet	11/8/1997 0:03	8.0					Disk
Zamama	Pr	18	176	E14	s0440873652	Violet	3/29/1998 0:37	3.0					Disk
Zamama	Pr	18	176	I24	s0520873452	Violet	10/11/1999 18:05	6.7					Terminator
Marduk	Pr	−23	211	G8	s0394552545	Violet	5/7/1997 18:41	11.4	51.0	20	52	20	Limb
Marduk	Pr	−23	211	E11	s0420773085	Violet	11/7/1997 21:19	8.2	76.0	8	76	8	Limb
Marduk	Pr	−25	212	C21	s0506405768	Violet	7/2/1999 4:01	2.9	32.9	10	92	28	Limb
Pillan	Pr	−11	242	C9	s0401863204	Violet	6/28/1997 2:40	6.2	99.3	20	109	22	Limb
Pillan	Pr	−12	243	E11	s0420789285	Violet	11/8/1997 0:03	8.0	105.4	25	118	28	Limb
Pele	Pe	−18	255	E4	s0374850023	Violet	12/20/1996 10:27	18.3	180.9	20	426	47	Limb
Pele	Pe	−18	255	G29	s0584334178	Violet	12/31/2000 8:23	10.3	297	30	299	30	Limb
Ra	Pr	−10	323	G1	s0350024288	Violet	6/29/1996 2:50	9.9	91.8	10	104	11	Limb

constraints, the Galileo SSI images were subjected to a lossy compression that produced blocky artifacts, particularly near the limb where plumes are most easily seen. The compression algorithm preserved the low spatial frequencies and the mean of each 8×8 pixel block, so the lossy compression does not affect the quantitative results presented below.

3.1. Dust plume sizes

Table 1 lists our inventory of Galileo images of dust plumes that were detected in daylight. Over the course of the 5 year mission, Galileo acquired 51 observations of sunlit plumes from 11 different volcanic centers. The best Galileo images of these plumes are presented in Fig. 1. In general, the Prometheus-type dust columns appeared similar to one another in size and shape but varied considerably in opacity and color. Fig. 1 shows that the Prometheus-type dust plumes displayed a characteristic fountain- or umbrella-shaped morphology with a dense central column,

contrasting with the shieldlike shapes of giant plumes that lack a dense core.

Kanehekili (Fig. 1A) was seen twice in 1997 in color observations taken during orbits G8 and E11. Plume sightings at Kanehekili are consistent with the record of surface changes (Geissler et al., 2004a), which suggest a reduced level of activity in the last half of the Galileo mission. Masubi (Fig. 1B) was not spotted until 1999, but was imaged at the highest spatial resolution of any of the Galileo era plumes. The images of Masubi from orbit C21 show a cone-shaped core with sharp margins, surrounded by a diffuse envelope of dust. Amirani (Fig. 1C) was active from late 1996 through 1999 but was only seen by Galileo from great distances. Thor (Fig. 1D) erupted unexpectedly late in the mission in 2001; the observations of Thor were intended to capture an expected eruption of Tvashtar. In contrast, Prometheus (Fig. 1E) was dependably present throughout the Galileo mission and has likely been erupting since the Voyager flybys. Zamama (Fig. 1F) was active during most of the mission, but was not detected in the tar-

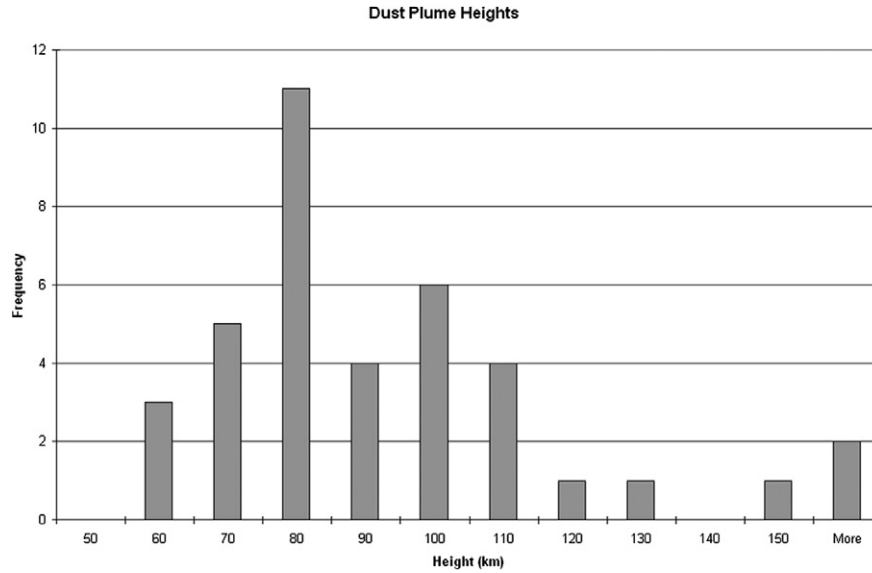


Fig. 2. Histogram of dust column heights. The frequency on the vertical axis refers to the number of observations, not the number of distinct plumes from different volcanic centers. Corrected heights of the Prometheus-type dust columns range from 52 km to 123 km, with a mean of 82 km and a standard deviation of 17 km. The outliers are Thor and Pele.

Table 2
Gas plume heights

Name	Type	Lat.	Lon.	Orbit	Image	Filter	GMT (M/D/Y H:M)	Res. (km/pix)	Meas. height (km)	Meas. error (km)	Corr. height (km)	Corr. error (km)	Position/notes
Grian	Pe	-12	13	C21	s0506570500	Clear	7/3/1999 7:46	30.2	339.8	45	506	67	Left limb
Kanehekili	Pr	-16	36	E11	s0420361523	Clear	11/4/1997 23:58	17.9	253.0	40	398	63	Left limb
Thor	Pr	39	131	I31	s0615325147	Violet	8/4/2001 22:56	18.3	440.0	30	444	30	Left limb; sunlit obs.
Thor	Pr	39	131	I31	s0615325547	Violet	8/4/2001 23:00	18.2	230.0	70	233	71	Left limb; sunlit obs.
Prometheus	Pr	-1	155	E15	s0449843800	Clear	5/31/1998 0:17	14.0	251.1	90	254	91	Right limb
Prometheus	Pr	-1	155	E15	s0449847913	Clear	5/31/1998 0:58	13.4	283.6	80	300	85	Right limb
Culann	Pr	-20	161	E15	s0449843800	Clear	5/31/1998 0:17	14.0	117.6	45	119	46	Right limb
Culann	Pr	-20	161	E15	s0449847913	Clear	5/31/1998 0:58	13.4	176.1	40	177	40	Right limb
Zamama	Pr	19	172	E15	s0449843800	Clear	5/31/1998 0:17	14.0	200.7	100	255	127	Right limb
Zamama	Pr	19	172	E15	s0449847913	Clear	5/31/1998 0:58	13.4	160.0	40	187	47	Right limb
Ra	Pr	-10	323	G1	s0350029700	Clear	6/30/1996 3:46	10.5	183.5	20	184	20	Left limb
Acala	Pr	12	331	E15	s0449843800	Clear	5/31/1998 0:17	14.0	230.2	50	246	53	Left limb
Acala	Pr	12	331	E15	s0449847913	Clear	5/31/1998 0:58	13.4	247.3	70	286	81	Left limb
Acala	Pr	10	335	G8	s0394394200	Clear	5/6/1997 16:00	18.6	313.9	60	349	67	Left limb
Acala	Pr	10	335	C9	s0401957745	Clear	6/28/1997 18:36	14.6	220.0	40			Disk
Acala	Pr	10	335	C10	s0413546765	Clear	9/18/1997 3:34	13.3	105.0	30	202	58	Left limb
Acala	Pr	10	335	E11	s0420858600	Clear	11/8/1997 11:44	13.7	204.3	65	243	77	Left limb
Culann/Prometheus				G8	s0394394200	Clear	5/6/1997 16:00	18.6	222.4	55			Right limb
Culann/Prometheus				E11	s0420858600	Clear	11/8/1997 11:44	13.7	205.4	40			Right limb
Zamama/Prometheus				E11	s0420361523	Clear	11/4/1997 23:58	17.9	127.0	60			Right limb

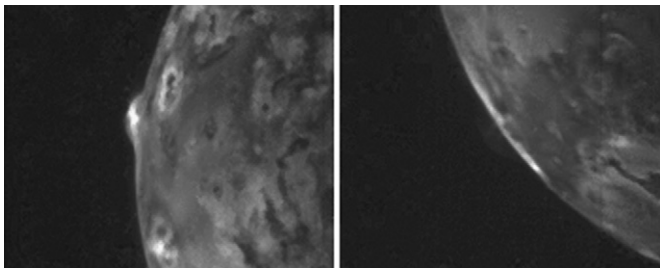


Fig. 3. Comparison of dust columns from Zamama (left) and Marduk (right) under nearly identical illumination and viewing conditions. The opacity of stealthy Marduk is an order of magnitude less than that of similarly sized Zamama.

be confidently associated with particular volcanic eruptive centers. Altogether, 16 observations of 8 different gas plumes met these criteria. The two observations of Thor were not taken in eclipse, but are included here because they are believed to show the extent

of the gas plumes, as explained in the next section. Three additional detections of plumes made up of gases from more than one volcanic center are included in Table 2 for completeness.

Fig. 4 shows the distribution of the measured heights of the gas plumes, corrected for foreshortening with the assumption that the location of the source was known in each case. The mean height of the gas jets of Prometheus-type plumes is found to be 258 ± 87 km, considerably larger than the optically thick dust columns. Significantly, we find no consistent difference between the gas plume heights of dusty plumes (such as Zamama or Prometheus) and stealth plumes (Acala and Culann). Acala is much brighter than other Prometheus-type plumes in eclipse, perhaps suggesting a higher gas density from this dust-free plume.

3.3. Contemporaneous gas and dust plume observations

A few Prometheus-type plumes were seen both in daylight and eclipse observations during the same Galileo orbit, giving us

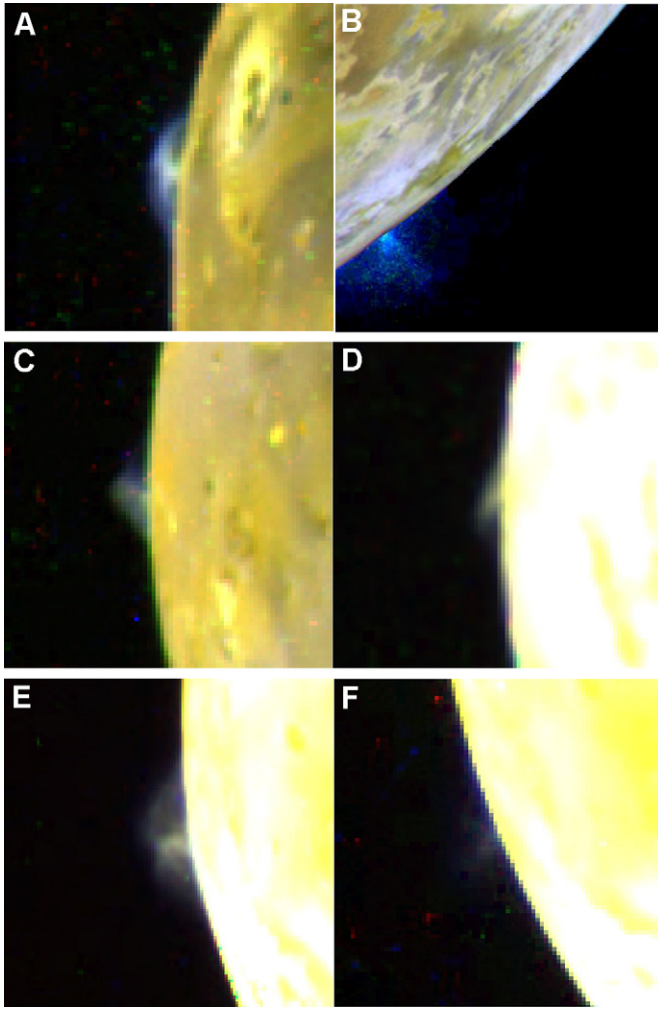


Fig. 11. Colors of Prometheus-type dust columns. (A) Zamama orbit E11. (B) Masubi orbit C21 (false color, Violet and 756 nm filters); PIA02502. (C) Prometheus orbit G8. (D) Prometheus orbit C9. (E) Pillan orbit C9. (F) Pillan orbit E11. Except for Masubi, the images are contrast-stretched linearly to show color differences. Prometheus and Pillan both appeared yellow during orbit C9, in contrast to the more typical blue hues of the plumes.

story. Most of the optically thick dust columns are blue in color, contrasting sharply with the yellows and reds of Io's surface. This indicates that the dust particles are smaller than or comparable in size to the wavelength of visible light, as explained in the next section. However, there are some notable exceptions to this general rule, and variations in color over time scales of weeks were noted even in observations of the same plumes.

Fig. 11 shows the variety of colors sported by Prometheus-type dust columns. Zamama (Fig. 11A) exemplifies the typical hue of the dust plumes; Zamama remained consistent in color throughout the Galileo observations, irrespective to first order of variations in incidence, emission, or phase angle. Masubi (Fig. 11B) was seen in false color (Violet and 756 nm filters) at a resolution in Violet of 3.2 km/pixel. Prometheus (Figs. 11C, 11D) appeared similar in color to Zamama except during orbit C9, when it was brighter than usual and yellow rather than blue, indicating coarser dust grains. Pillan (Figs. 11E, 11F) was yellow when it first erupted but reverted to a more typical blue in the waning stages of the eruption. Pillan's dense plume produced an unusual black deposit on the surface, obscuring a portion of Pele's red ring. This indicates that silicates were entrained in Pillan's dust plume. One other dubious detection of an unusually colored plume was that of Grian in

orbit C10. A faint smudge was seen near the limb of Io in the Red and 756 nm filter images but was not visible in the sharper Violet and Green images. It is possible that this observation represents a conspiracy of compression artifacts, but it is included in Table 1 because a giant red ring erupted from Grian two years later during orbit C21, and we cannot rule out the possibility that red dust was exhaled from the volcano as a precursor to the later explosion.

Fig. 12 shows our measurements of the spectra of individual plumes, in units of reflectance (I/F). There is a great deal of variability in plume brightness and color, but in general the reflectance of the optically thick dust columns tends to be higher at shorter wavelengths. Fig. 13 shows the mean color of the most reliable dust column observations, excluding the exceptional yellow and red plumes discussed earlier. Here we have normalized the measurements to the Violet filter reflectance, to compensate for brightness variations from one observation to another, and then averaged the resulting ratios. This is the spectrum of "typical" plumes that we will seek to match with theoretical calculations in the next section. One additional observation of importance is that there is little apparent variation of color with phase angle, as shown in Fig. 14.

3.8. Summary of observations

Galileo observations of volcanic plumes on Io reveal a zoo of individuals with different sizes, morphologies, colors, and opacities. The following features must be explained by any comprehensive theory that seeks to explain the diversity of Io's explosive eruptions: (1) the optically thick dust columns of Prometheus-type plume decouple from the gas flow and fall short of the extent of the gas plumes that drive them (gas plumes are ~ 3 times larger than the optically thick dust plumes); (2) refractory materials like silicates can sometimes be entrained in the dust from Prometheus-type plumes; (3) the dust loads of Prometheus-type plumes are independent of plume height (initial gas temperature), as expected for entrained particles; (4) the visible surface changes produced by Prometheus-type plumes are consistent with the trajectory of the dust, and not that of the gas; (5) a fine-grained population of particles that follows the gas flow can be seen in long exposure images of certain Prometheus-type plumes like Thor; (6) Pele-type giant plumes have similarly sized dust columns, gas plumes and surface deposits, implying a lack of separation of the gas and dust components; and (7) Pele-type plumes lack a distinct central core, consistent with the possibility that the dust particles are condensed directly from the gas.

4. Interpretation

To estimate particle sizes and plume masses, we used the publicly available program Mieplot by Philip Laven (www.philiplaven.com). Mieplot provides a choice of Mie, Rayleigh, and other scattering laws and outputs predicted intensity as a function of wavelength and optical power as a function of scattering angle. It allows for a distribution of particle sizes, which is important because of the strong wavelength- and scattering-angle-dependence predicted for particular particle sizes. For our simulations, we assumed a normal distribution of dust grain radii with a standard deviation of 20% about the mean. The modeling results reported here are most appropriate to liquid droplets; crystals and complex aggregates such as snowflakes are not well modeled by spherical scatterers. We assumed the refractive index of liquid SO₂ (1.34; Musso et al., 2000) but the conclusions are little changed over a range of refractive indices from 1.3 to 1.9 (appropriate to H₂O and S, respectively).

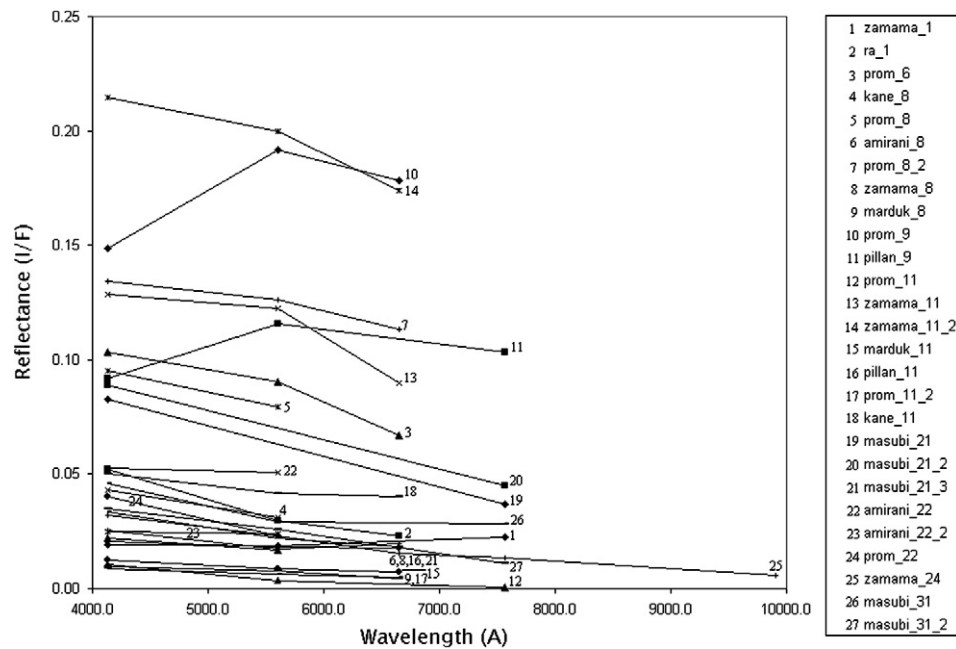


Fig. 12. Spectra of Prometheus-type dust plumes, in units of reflectance (I/F). There is a great deal of variability in plume brightness and color, but in general the reflectance of the optically thick dust columns is higher at shorter wavelengths. Notable exceptions are Prometheus and Pillan, which both appeared yellow during orbit C9 (cf. Fig. 11).

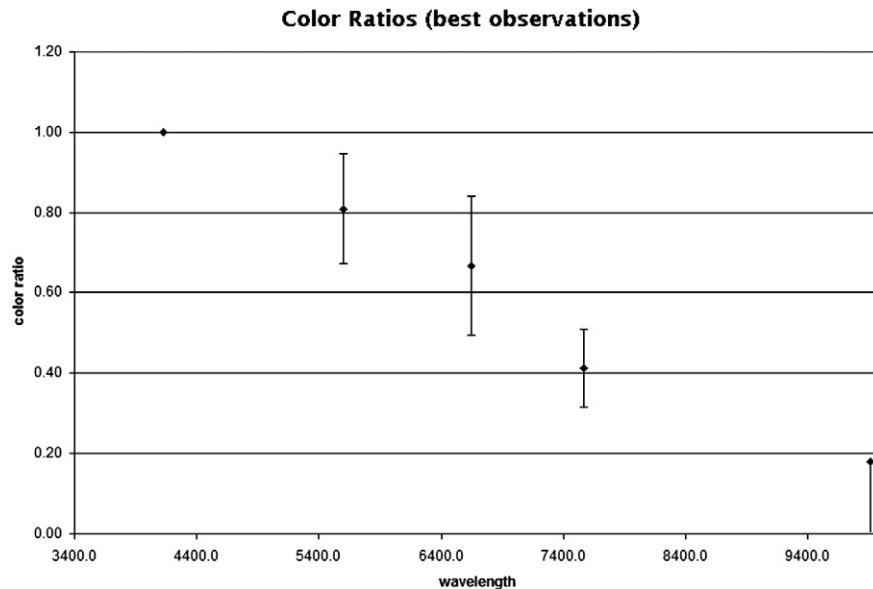


Fig. 13. Mean color of Prometheus-type dust plumes. This figure shows the spectrum of the optically thick dust columns that were best observed by Galileo's camera, excluding the exceptionally colored red and yellow plumes. The measurements were normalized to the Violet filter reflectance, to compensate for brightness variations from one observation to another, and the resulting ratios were then averaged. The point at 9680 Å is an upper limit.

4.1. Dust particle sizes

Particles that are very small or very large in comparison to the wavelength are inefficient scatterers of sunlight, so the visibility of plumes is strongly affected by the wavelength of the observations. Moreover, the estimated mass of plumes depends sensitively on the grain size assumed. We used the plume color measurements described previously in Section 3.7 to establish the mean grain size of the optically thick dust plumes. Both Rayleigh and Mie theory predict a λ^{-4} dependence of scattered light intensity for particles with radii $r \ll \lambda$. As we have seen, the plume colors recorded by Galileo SSI display an approximately linear decrease in intensity over the wavelength interval from 410 nm to 756 nm. These observations rule out very small particles (1 nm to 10 nm)

as the dominant scatterers in the optically thick dust columns. Instead, the simulations suggest that scattering in the dust plumes is dominated by particles that are smaller than, but comparable to, the wavelength of visible light. The best fits to the Galileo color observations are obtained with particle radii in the range 80 nm to 120 nm, depending on the scattering angle (Fig. 15). The model fits are imperfect, however; theory predicts some variations in color with scattering angle that are not seen in the observational data of Fig. 14. We conclude from these simulations that (1) nanometer-sized particles do not dominate the scattering of visible light by the optically thick dust columns; (2) the dust grains that contribute most to the scattering seen by Galileo SSI are smaller than but comparable to the wavelength of visible light (i.e., in the radius range of 80 nm to 120 nm); and (3) the distri-

fur on Io? The data from Voyager, Cassini, and New Horizons will be needed to discern the nature of the giant plumes and their role in Io's resurfacing.

6. Conclusions

Our study has shown that there is more to Io's mysterious plumes than meets the eye, found answers to some of the questions posed earlier, and raised new questions to be addressed by other observations and data sets. The main findings of this work can be summarized as follows.

1. *Source of the dust particles.* Prometheus-type plumes possess two distinct dust components. The first component is coarse grained "ash" that is evidently entrained with the gas upon eruption but falls short of the gas flow, is optically thick at visible wavelengths (indicating particles that are comparable in size to the wavelength of visible light), varies in number density independently of the gas column height (i.e., the temperature of the gas), and may include refractory components such as silicates. The second component is fine grained dust spotted in some Prometheus-type plumes such as Thor and Loki that is small enough ($r < 10$ nm) to be carried along with the gas flow, and could represent SO₂ "snowflakes" condensed from the gas phase.
2. *Cause of the surface changes.* The visible surface changes produced by Prometheus-type plumes on Io are caused by the coarse grained ash columns that are optically thick at visible wavelengths. This result applies strictly to the plume deposits documented in SSI images by Geissler et al. (2004a); it does not rule out surface condensation from the gas phase or deposition of fine particles that may produce frosts of SO₂ that are transparent at visible wavelengths.
3. *Mass of plume dust.* The mass of coarse grained ash entrained by an individual Prometheus-type plume is typically 10^6 kg, implying ash production rates of $\sim 10^3$ kg s⁻¹ per plume. The ash comprises only <10% of the mass of the gas. In contrast, the fine dust suspended by Thor comprises at least 10^8 kg, comparable to the mass of the gas (assuming that Thor's gas column density was equal to that of Prometheus).
4. *Significance for resurfacing.* Despite their spectacular appearance, the optically thick ash columns from Prometheus-type plumes are responsible for only a small fraction of the resurfacing of the satellite. The erasure of impact craters must be achieved either by the emplacement of lava flows or by more subtle deposition from the plumes that was missed by Galileo's camera, such as SO₂ snowfall or direct condensation from the gas phase onto the surface. More analysis of observations from instruments other than Galileo SSI are needed to quantify the importance of these alternate deposition mechanisms.
5. *Giant plumes.* Limited information about the Pele-type plumes from Galileo SSI suggests that the gas plumes, dust columns, and surface changes are all similarly sized, indicating that the gas and dust components of giant plumes do not separate. The poor visibility of Pele's dust is another indication that the particles must be much smaller than those of the dust columns from Prometheus-type plumes. Together, these facts admit the possibility that it snows (or rains) sulfur from Io's giant plumes.

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